

# Jacob (Jake) Carlson

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<https://jscarlson.github.io>

## EDUCATION

**Harvard University**, Cambridge, Massachusetts  
Ph.D. Economics, August 2022 – May 2027 (expected)

**Yale University**, New Haven, Connecticut  
B.A. Statistics and Data Science, August 2016 – May 2020  
B.A. Ethics, Politics, and Economics, August 2016 – May 2020  
*Summa cum laude* with distinction in both subjects

## HONORS AND AWARDS

- PhD Fellowship Finalist, Two Sigma
- Presidential Scholar, Harvard University
- Phi Beta Kappa, Yale University
- Undergraduate Thesis Prize Winner, Forethought Foundation for Global Priorities Research

## WORKING PAPERS

- Carlson, Jacob and Neil Shephard. “When are time series predictions causal? The potential system and dynamic causal effects.” Working paper (2026).
- Carlson, Jacob. “Making Interpretable Discoveries from Unstructured Data: A High-Dimensional Multiple Hypothesis Testing Approach.” Working paper (2025).
- Carlson, Jacob and Melissa Dell. “A Unifying Framework for Robust and Efficient Inference with Unstructured Data.” Working paper (2025).
- Andrews, Isaiah, Harvey Barnhard, and Jacob Carlson. “True and Pseudo-True Parameters.” Working paper (2024). Conditionally accepted, *Econometric Theory*.

## CONFERENCE PUBLICATIONS

- Carlson, Jacob, Tom Bryan, and Melissa Dell. “Efficient OCR for Building a Diverse Digital History.” *ACL* (2024). **Oral presentation.**
- Dell, Melissa, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D’Amico-Wong, Quan Le, Pablo Querubin, Leander Heldring. “American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers.” *NeurIPS Datasets and Benchmarks Track* (2023).
- Bryan, Tom, Jacob Carlson, Abhishek Arora, Melissa Dell. “EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge.” *EMNLP System Demonstrations* (2023).
- Shen, Zejiang, Ruochen Zhang, Melissa Dell, Benjamin Lee, Jacob Carlson, and Weining Li. “LayoutParser: A Unified Toolkit for Deep Learning Based Document Image Analysis.” *ICDAR* (2021). **Oral presentation.**

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| <b>JOURNAL PUBLICATIONS</b> | <ul style="list-style-type: none"> <li>• Carlson, Jacob, Trevor Incerti, and P.M. Aronow. “Dyadic Clustering in International Relations.” <i>Political Analysis</i> (2024).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>EMPLOYMENT</b>           | <p><b>Harvard University</b>, Cambridge, Massachusetts<br/> Data Science Fellow in Deep Learning                      September 2020 – May 2022</p> <ul style="list-style-type: none"> <li>• Developed deep learning methods and tools for social science data curation with Melissa Dell</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>TEACHING</b>             | <p><b>Harvard University</b>, Cambridge, Massachusetts</p> <ul style="list-style-type: none"> <li>• <i>Causal Inference in Economics</i> (ECON 2148, Graduate): Spring 2025, 2026</li> <li>• <i>Time Series</i> (STAT 242, Graduate): Fall 2024, 2025</li> <li>• <i>Unleashing Novel Data at Scale</i> (ECON 2355, Graduate): Spring 2021</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>INVITED TALKS</b>        | <ul style="list-style-type: none"> <li>• <b>2026:</b> ESIF Economics and AI+ML Meeting (Cornell, June); AI and Economics Summer Conference (UChicago Booth, August)</li> <li>• <b>2022:</b> ASSA Annual Meeting (Virtual, January)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>SOFTWARE</b>             | <ul style="list-style-type: none"> <li>• hdmht-discovery: <a href="https://github.com/jscarlson/hdmht-discovery">https://github.com/jscarlson/hdmht-discovery</a></li> <li>• mar-s (with Melissa Dell): <a href="https://github.com/jscarlson/mar-s">https://github.com/jscarlson/mar-s</a></li> <li>• NetworkTargeting (with Davide Viviano): <a href="https://github.com/dviviano/NetworkTargeting">https://github.com/dviviano/NetworkTargeting</a></li> <li>• EffOCR (with Tom Bryan and Melissa Dell): <a href="https://pypi.org/project/efficient-ocr">https://pypi.org/project/efficient-ocr</a></li> <li>• dcr (with Trevor Incerti and P.M. Aronow): <a href="https://github.com/jscarlson/dcr">https://github.com/jscarlson/dcr</a>, <a href="https://github.com/jscarlson/stata-dcr">https://github.com/jscarlson/stata-dcr</a></li> </ul> |
| <b>ACADEMIC SERVICE</b>     | <p><i>Referee:</i> Journal of Political Economy</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>LAST UPDATED</b>         | <p>August 31, 2026</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |